

# Department of Computer Science and Engineering (CSE) BRAC University

## Lecture 3

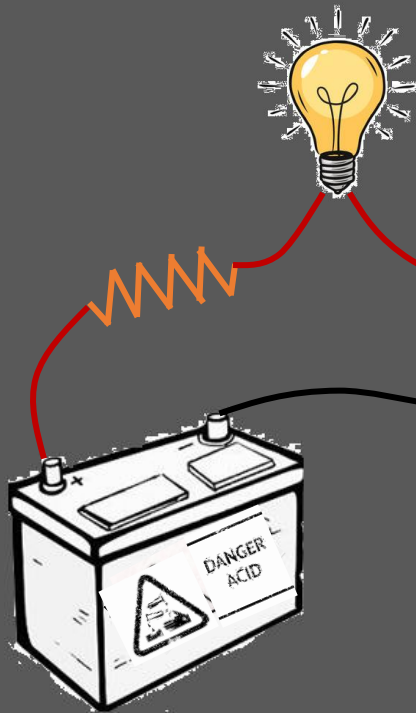
### CSE250 - Circuits and Electronics

# KCL AND KVL



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# Course Outline: broad themes



**Circuit Laws**

Kirchhoff's  
Current  
Law



Methods of  
Analysis

Circuit  
Theorems

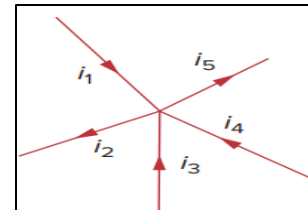
First Order  
Circuits

AC Circuits

# Kirchhoff's Current Law (KCL)

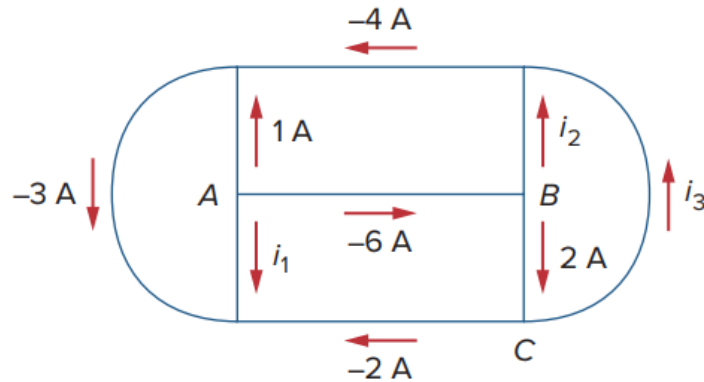
- **Kirchhoff's current law (KCL)** the algebraic sum of the currents entering a node is equal to the algebraic sum of the currents leaving the node.
- Mathematically,  $\sum_{n=1}^N i_n = 0$ , where  $N$  is the number of branches connected to the node and  $i_n$  is the  $n$ th current entering (or leaving) the node.
- Assume a set of currents  $i_k(t)$ ,  $k = 1, 2, \dots$ , flow into a node. The algebraic sum of currents at the node is,  $i_{total}(t) = i_1(t) + i_2(t) + i_3(t) + \dots$
- Integrating both sides,  $q_{total}(t) = q_1(t) + q_2(t) + q_3(t) + \dots$ ,  $[q_k(t) = \int i_k(t) dt]$
- The *law of conservation of electric charge* requires that the algebraic sum of electric charges at the node must not change; that is, the node stores no net charge. Thus,  $q_{Total}(t) = 0 \rightarrow i_T(t) = 0$ , confirming the validity of KCL.
- For the adjacent node,  $i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$

$$\text{or, } i_1 + i_3 + i_4 = i_2 + i_5$$



# Example 1

(i) Find  $i_1$ ,  $i_2$ , and  $i_3$



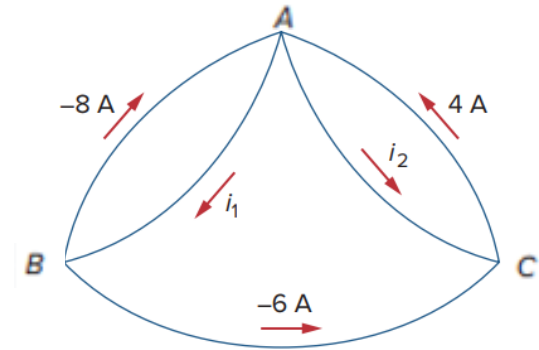
Note that, in both the circuits  $A, B$ , and  $C$  are the same nodes. It is more appropriate to call them junctions in this case.

KCL at junction A,  
 $i_1 + 1 + (-6) = 0$   
 $\Rightarrow i_1 = 5 \text{ A}$

KCL at junction B,  
 $i_2 + 2 = -6$   
 $\Rightarrow i_2 = -8 \text{ A}$

KCL at junction C,  
 $2 = (-2) + i_3$   
 $\Rightarrow i_3 = 4 \text{ A}$

(ii) Find  $i_1$  and  $i_2$

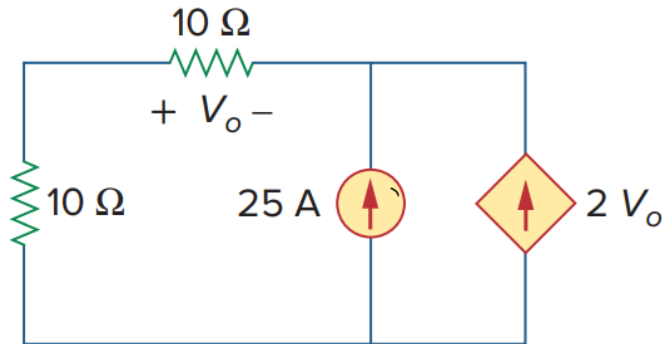


KCL at junction B,  
 $i_1 = (-8) + (-6)$   
 $\Rightarrow i_1 = -14 \text{ A}$

KCL at junction C,  
 $i_2 + (-6) = 4$   
 $\Rightarrow i_2 = 10 \text{ A}$

# Example 2

- Find  $V_0$  and power absorbed/supplied by the dependent source with appropriate  $\pm$  sign.



Current through the series resistances =  $25 + 2V_0$

According to the Ohm's law,

$$V_0 = -10 \times (25 + 2V_0)$$

$$V_0 = -11.9 \text{ V}$$

The voltage across the dependent source is,

$$V_x = (10 + 10) \times (25 + 2V_0) = 24 \text{ V}$$

With the polarity of  $V_x$  and the direction of the current ( $2V_0$ ) given, according to the passive sign convention, the dependent source is supplying power. So,

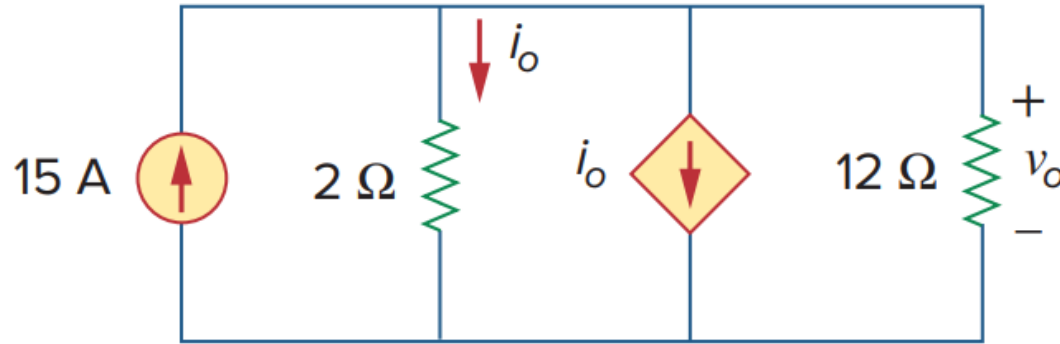
$$p = -24 \times 2V_0 = 571.2 \text{ W}$$

The power is positive, hence, the dependent source is actually absorbing power. This is true as  $V_0$  is negative, the current  $2V_0$  is actually flowing in the opposite direction.



# Problem 1

- Determine  $v_0$  and  $i_0$  using KCL and Ohm's law.

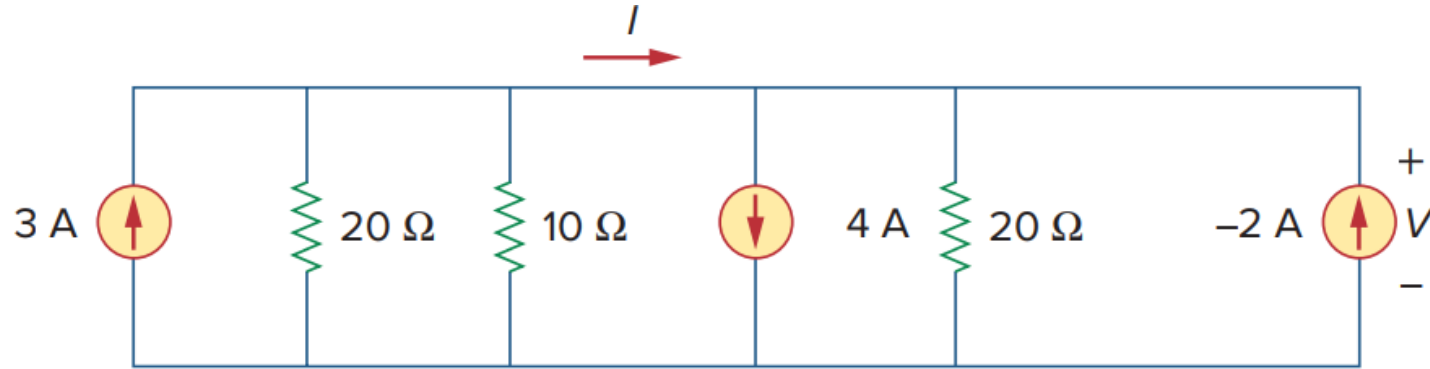


Ans:  $v_0 = 13.85\text{ V}$ ;  $i_0 = 6.92\text{ A}$ .



# Problem 2

- Find the  $V$  and  $I$  shown in the following circuit using KCL and Ohm's law.

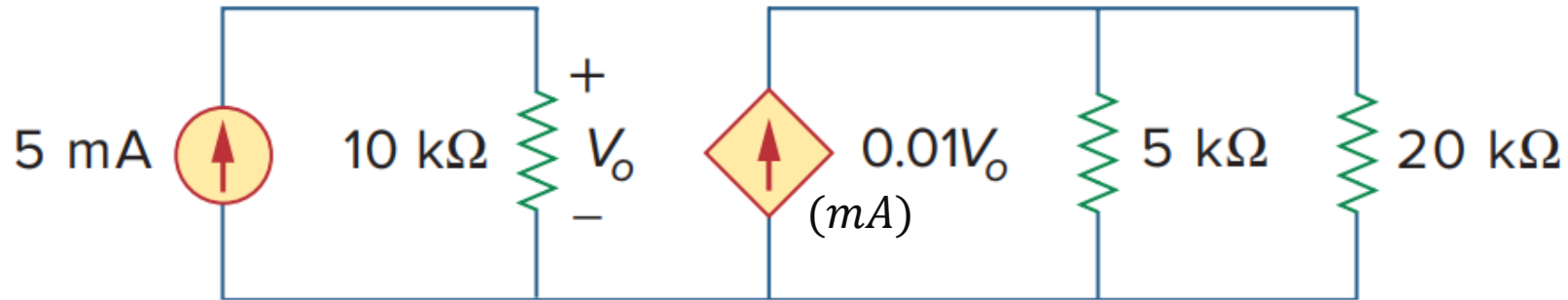


Ans:  $V = -15\text{ V}$ ;  $I = 5.25\text{ A}$ .



# Problem 3

- For the network shown below, find the current, voltage, and power associated with the  $20\text{ k}\Omega$  resistor using KCL and Ohm's law.

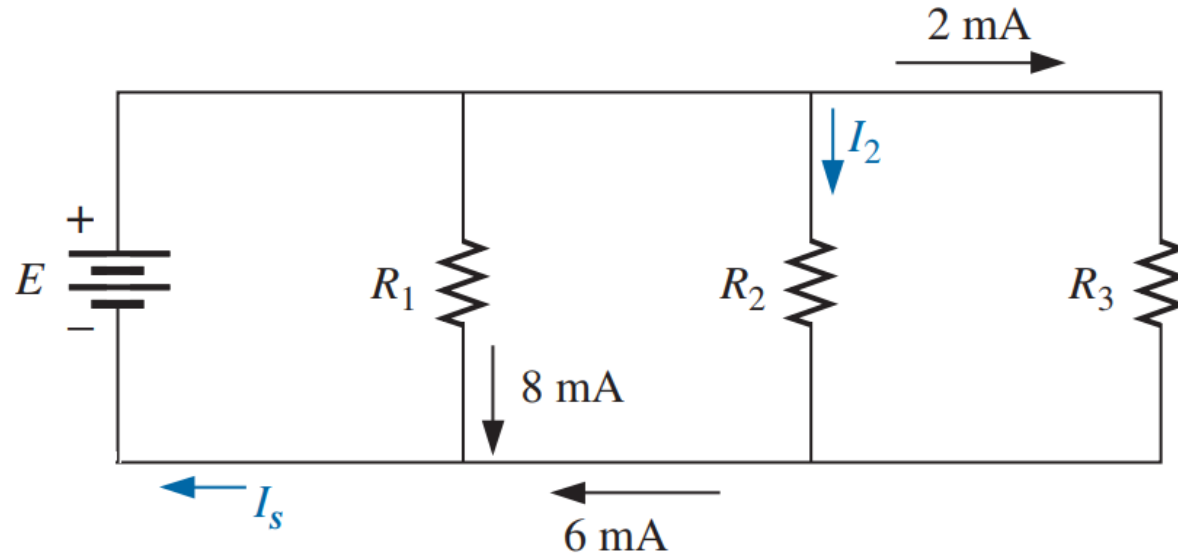


Ans:  **$0.1\text{ mA}$ ,  $2\text{ V}$ ,  $0.2\text{ mW}$**



# Problem 4

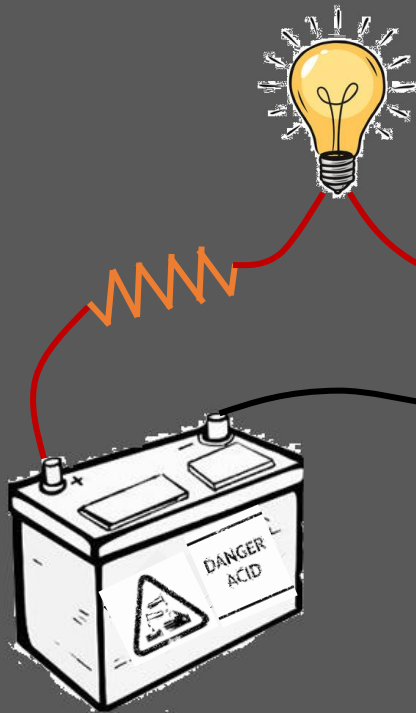
- Using KCL, determine the unknown currents.



$$\text{Ans: } I_2 = 4\text{ mA}, I_s = 14\text{ mA}$$



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Circuit Laws

Kirchhoff's  
Voltage  
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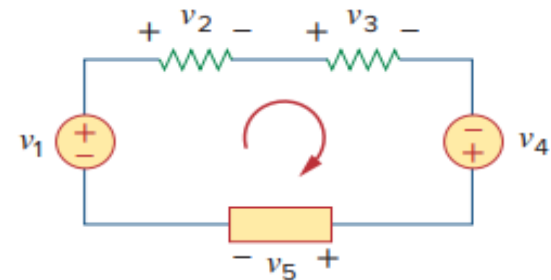
# Kirchhoff's Voltage Law (KVL)

- Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a path (or loop) is zero. KVL can be applied both in loop or path consisting of open circuits.
- Mathematically,  $\sum_{m=1}^M v_m = 0$ , where  $M$  is the number of voltages (or branches) in the loop and  $v_m$  is the  $m^{th}$  voltage.
- To illustrate KVL, consider the circuit shown. The sign on each voltage is the polarity of the terminal encountered first as we travel around the loop.
- If we start with the voltage source and go clockwise around the loop as shown; then voltages would be  $-v_1$ ,  $+v_2$ ,  $+v_3$ ,  $-v_4$ , and  $+v_5$ , in that order. For example, as we reach branch 3, the positive terminal is met first; hence, we have  $+v_3$ . For branch 4, we reach the negative terminal first; hence,  $+v_4$ . Thus, KVL yields

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

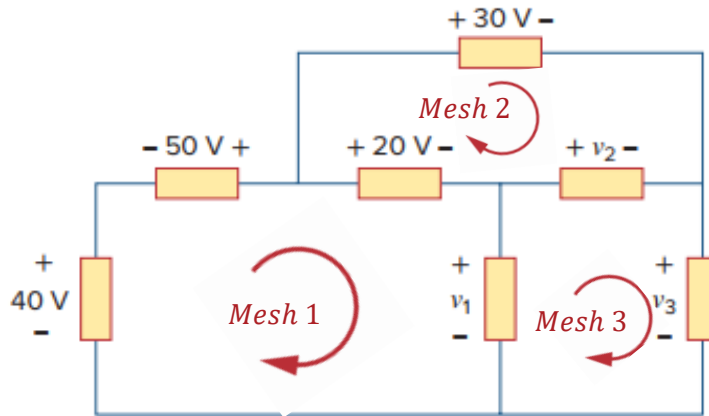
$$\text{or, } v_2 + v_3 + v_5 = v_1 + v_4$$

Sum of voltage drops = Sum of voltage rises



# Example 3

- Determine  $v_1$ ,  $v_2$ , and  $v_3$  using KVL.



KVL at mesh 1,

$$-40 - 50 + 20 + v_1 = 0$$

$$v_1 = 70 \text{ V}$$

KVL at mesh 2,

$$-20 + 30 - v_2 = 0$$

$$v_2 = 10 \text{ V}$$

KVL at mesh 3,

$$-v_1 + v_2 + v_3 = 0$$

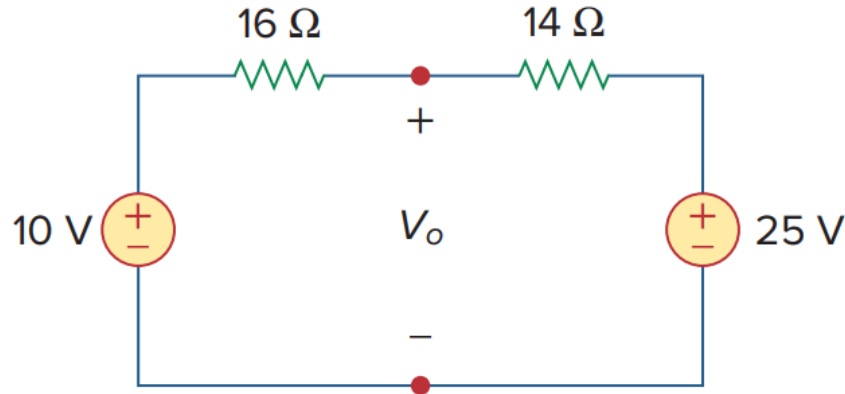
$$-70 + 10 + v_3 = 0$$

$$v_3 = 60 \text{ V}$$



# Example 4

- Determine  $V_0$  using KVL and Ohm's law.



Let's assume that the current through the series circuit is  $i$ .

Applying KVL around the loop,

$$-10 + 16i + 14i + 25 = 0$$

$$i = -0.5 \text{ A}$$

$V_0$  can be found either by applying KVL through the loop consisting of  $V_0$ , 14 Ω, and 25 V or applying KVL through the loop consisting of  $V_0$ , 16 Ω, and 10 V. That is,

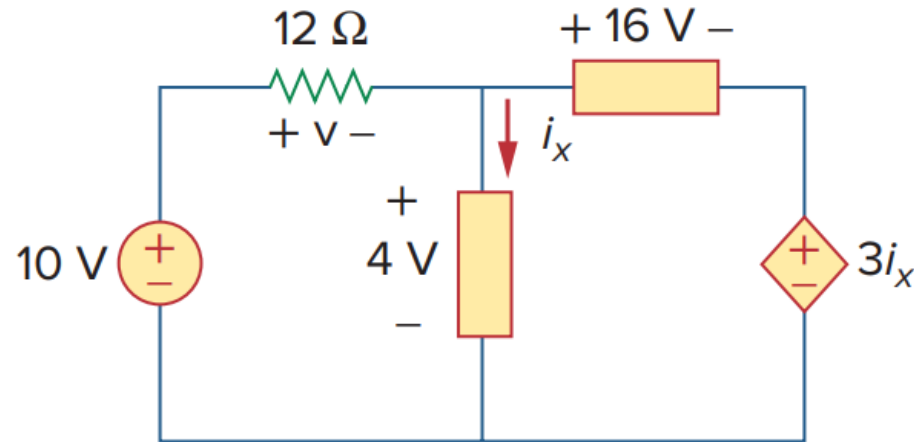
$$-V_0 + 14i + 25 = 0, \text{ or } V_0 = 18 \text{ V}$$

Or,

$$-10 + 16i + V_0 = 0, \text{ or } V_0 = 18 \text{ V}$$

# Problem 5

- Find  $v$  and  $i_x$  in the following circuit using KVL.

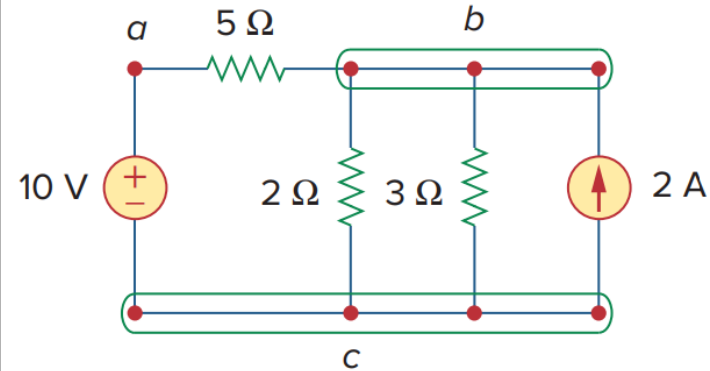


Ans:  $v = 6\text{ V}$ ;  $i_x = -4\text{ A}$ .



# # of Solvable KVL Equations

- In general, a circuit with  $n$  nodes and  $b$  branches will have  $b - n + 1$  *independent KVL equations*.
- It is necessary to apply KVL only to these loops, which will in total, traverse each branch at least once in the process. For example, in the adjacent circuit, the number of independent KVL equations will be  $5 - 3 + 1$  or 3. However, not any 3 equations will be solvable.
- For example, if 3 KVL equations are formed from the 3 loops: (i) 10 V, 5  $\Omega$ , 2  $\Omega$ , (ii) 10 V, 5  $\Omega$ , 3  $\Omega$ , and (iii) 2  $\Omega$ , 3  $\Omega$ , there will be no finite set of solution as the 2 A branch is not traversed in any of the three equations.
- They are not independent as any of the 3 equations will be a linear combination of the other two.

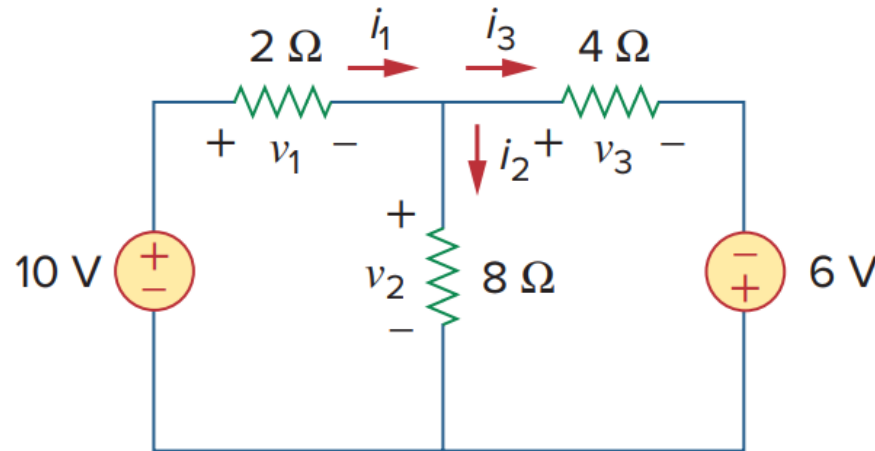


- Try applying KVL to every mesh first.
- Be careful not to allow the number of variables in the equations exceed the number of meshes.



# Problem 6

- Following only the loops, how many KVL equations can be formed from the circuit shown below? Form all the KVL equations in terms of  $v_1$ ,  $v_2$ , and  $v_3$ . Are the simultaneous equations solvable? Why? Determine all the voltages and currents labelled.

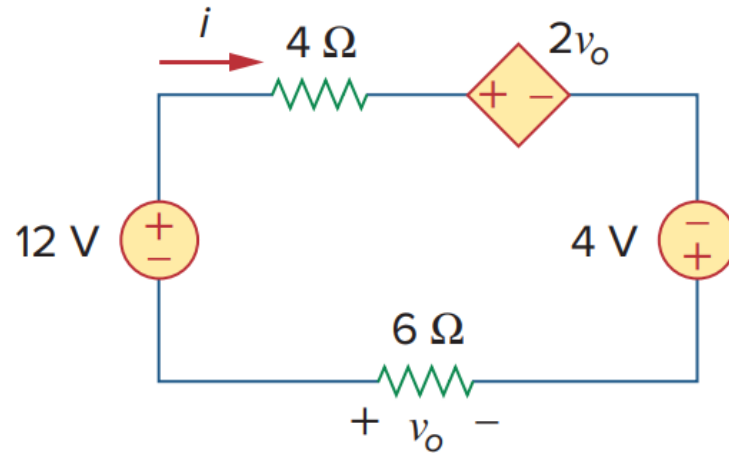


Ans:  $v_1 = 6\text{ V}$ ;  $v_2 = 4\text{ V}$ ;  $v_3 = 10\text{ V}$ .  
 $i_1 = 3\text{ A}$ ;  $i_2 = 0.5\text{ A}$ ;  $i_3 = 2.5\text{ A}$



# Problem 7

- Find  $v_o$  and  $i$  in the circuit using KVL and Ohm's law.

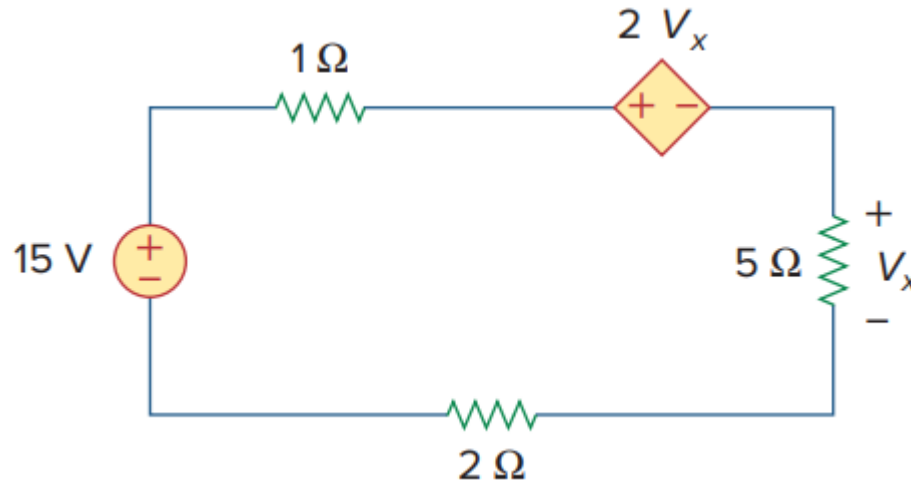


Ans:  $v_o = 48\text{ V}$ ;  $I = -8\text{ A}$ .



# Problem 8

- Find  $V_x$  using KVL and Ohm's law.

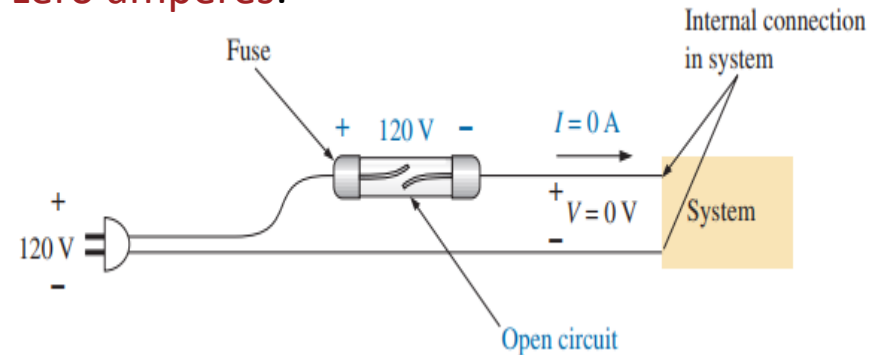
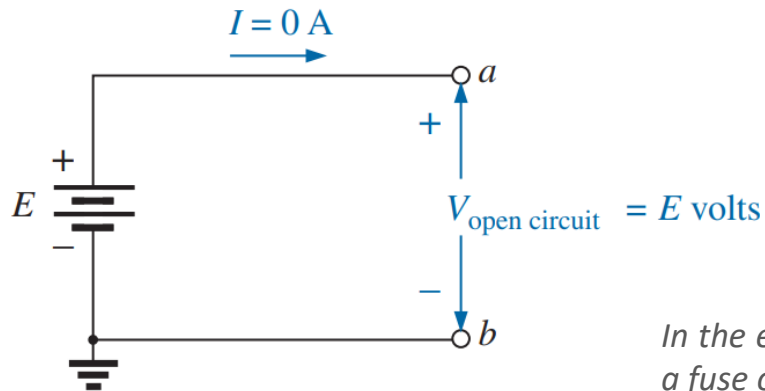


Ans:  $V_x = 4.167\text{ V}$ .



# Open circuit

- An *open circuit* is two isolated terminals not connected by an element of any kind. It is the limiting case of a resistor where the resistance approaches infinite.
- Any element with  $R \rightarrow \infty$  is an open circuit.  $i = 0 = \lim_{R \rightarrow \infty} \frac{v}{R}$
- Indicating that, an open circuit can have a potential difference (voltage) across its terminals, but the **current is always zero amperes**.

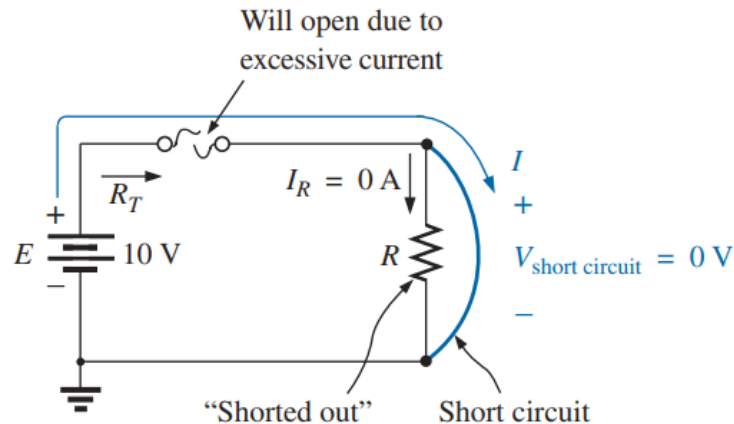
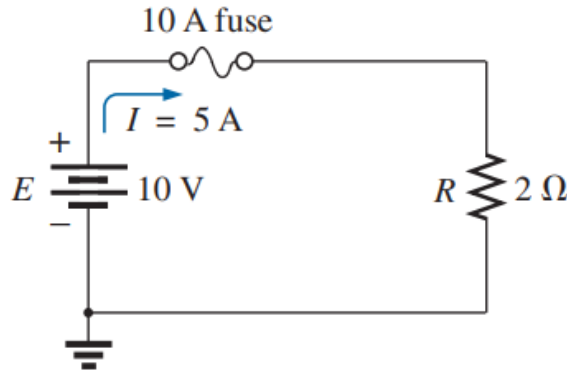


*In the event of an excessive current flow, a fuse opens to protect appliances.*



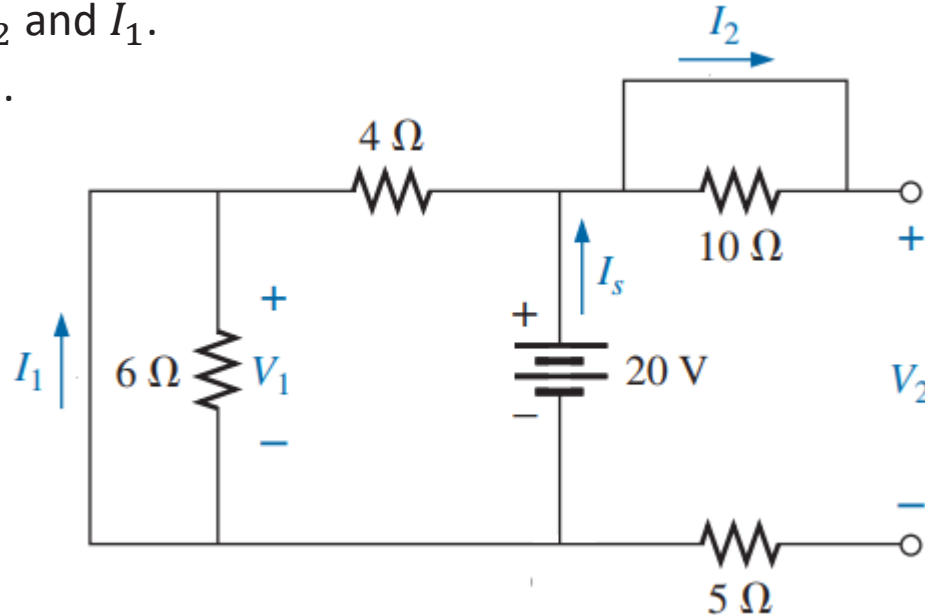
# Short circuit

- A *short circuit* is a very low resistance, direct connection between two terminals of a network. It is the limiting case of a resistor where the resistance approaches zero.
- Any element with  $R = 0$  is a short circuit.  $v = 0 = \lim_{R \rightarrow 0} iR$
- Indicating that, a short circuit can carry a current of a level determined by the external circuit, but the **potential difference (voltage) across its terminals is always zero volts.**



# Problem 9

- What are the values of
  - $V_1$  and  $I_2$ .
  - $V_2$  and  $I_1$ .
  - $I_s$ .

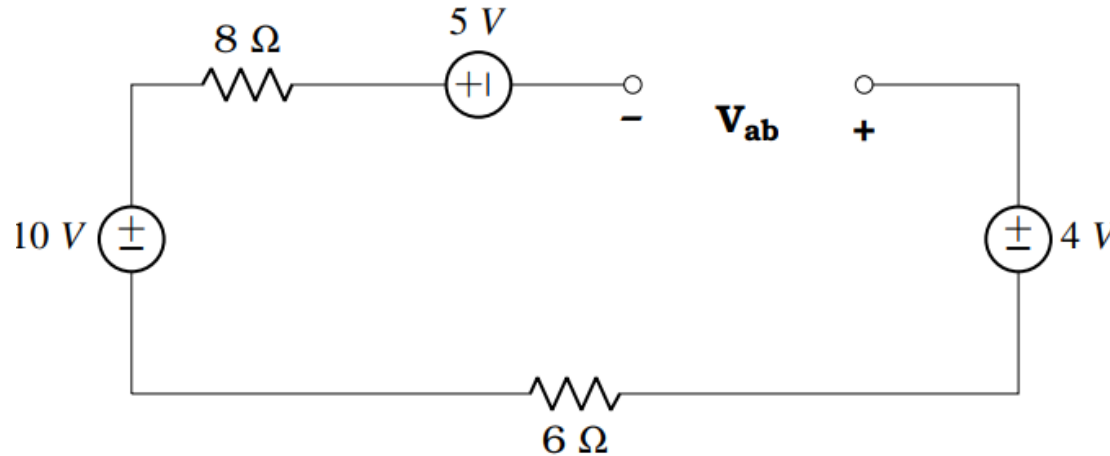


Ans:  
 a.  $0\text{ V}, 0\text{ A}$   
 b.  $20\text{ V}, -5\text{ A}$   
 c.  $5\text{ A}$



# Problem 10

- Determine the voltage  $V_{ab}$  as indicated.

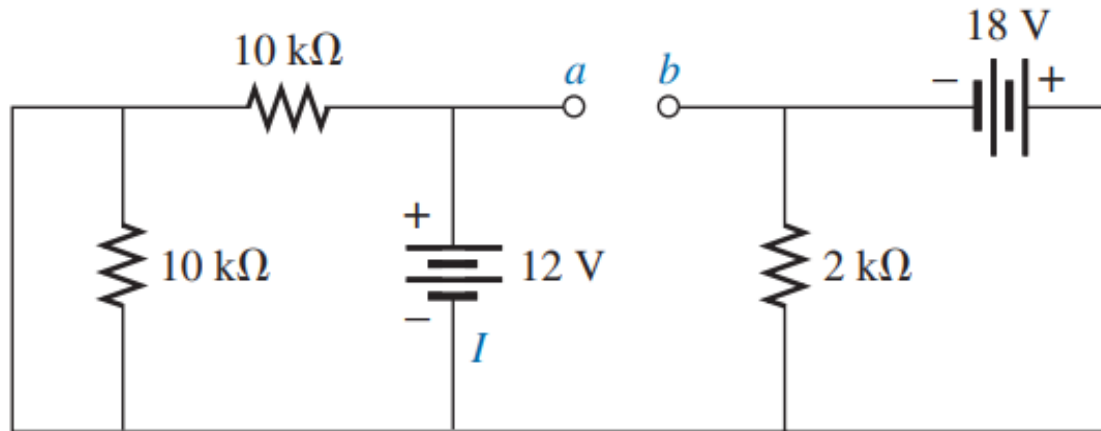


Ans:  $V_{ab} = -1\text{ V}$ .



# Problem 11

- For the network shown below, determine the voltage between terminals  $a$  and  $b$  and the current  $I$  supplied by the battery.

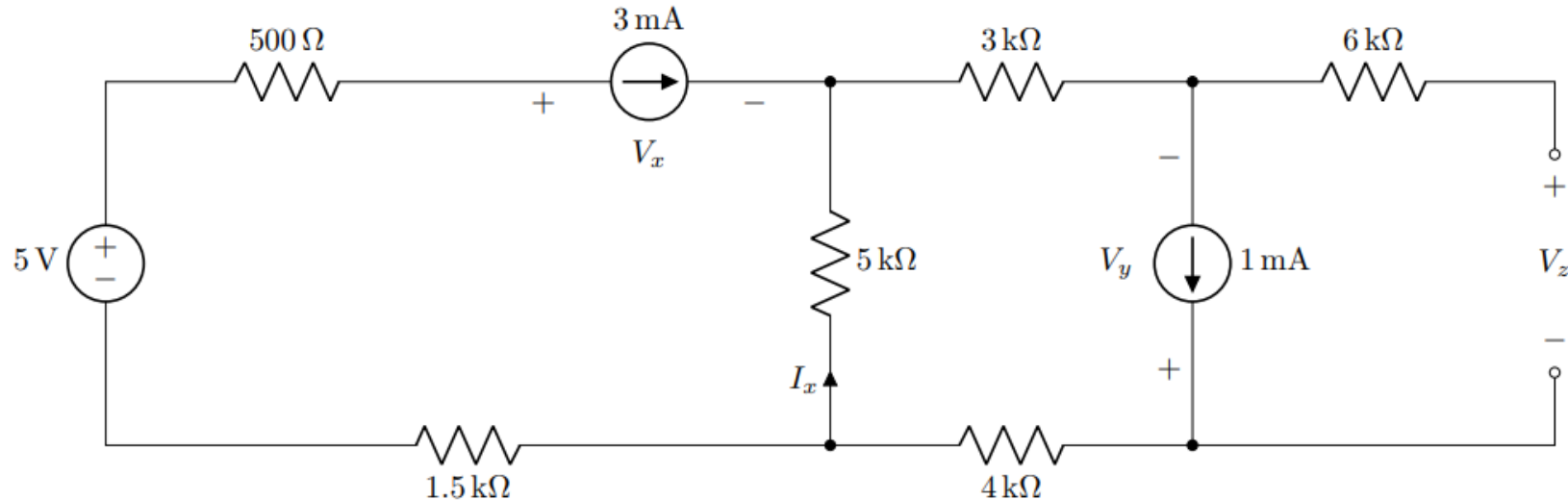


Ans:  $V_{ab} = 30\text{ V}; I = 1.2\text{ mA}$ .



# Problem 12

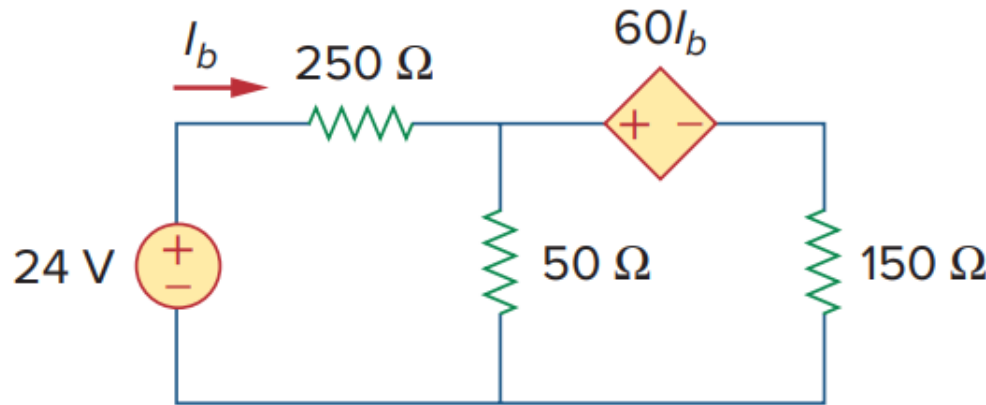
- Using KCL, KVL, and Ohm's law only, determine  $I_x$ ,  $V_x$ ,  $V_y$ , and  $V_z$ .



$$\text{Ans: } I_x = -2 \text{ mA}; V_x = -11 \text{ V}; V_y = -7 \text{ V}; V_z = 7 \text{ V}$$

# Problem 13

- Using circuit laws, determine the power of the dependent source.

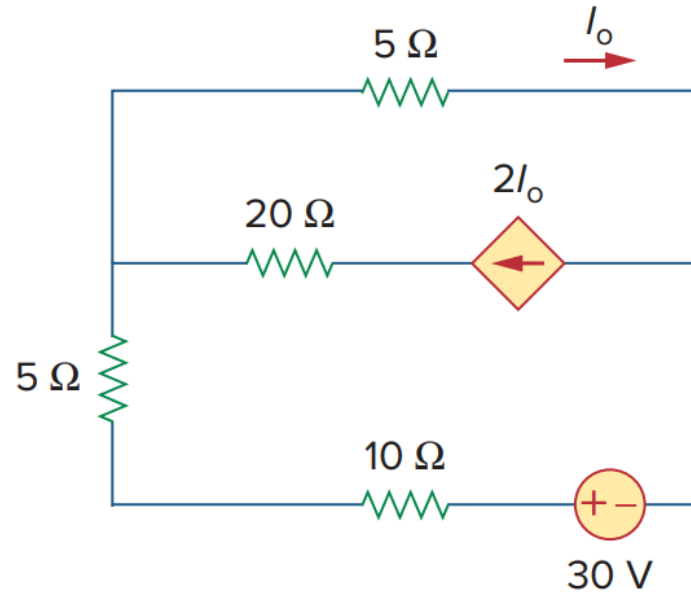


Ans:  $P_{60I_b} = -240 \text{ mW}$



# Problem 14

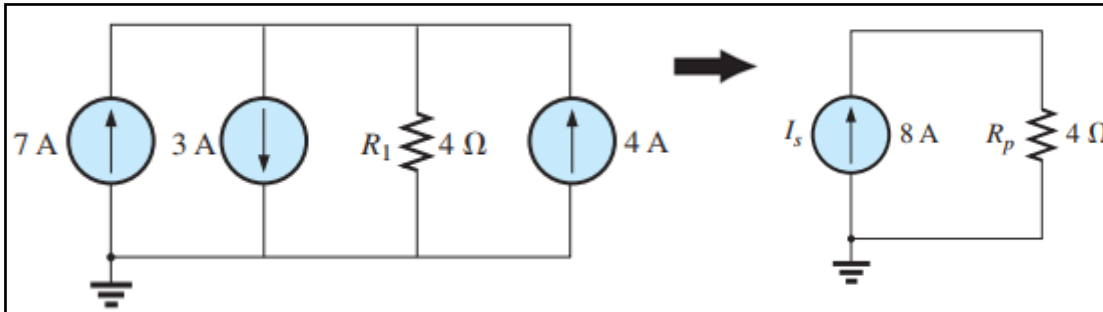
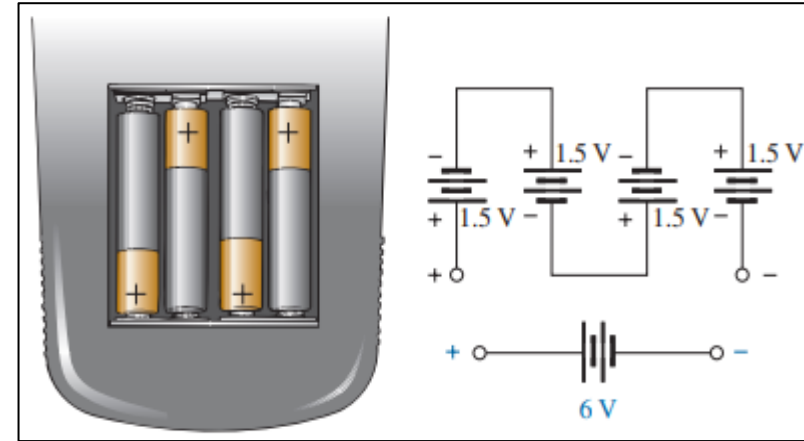
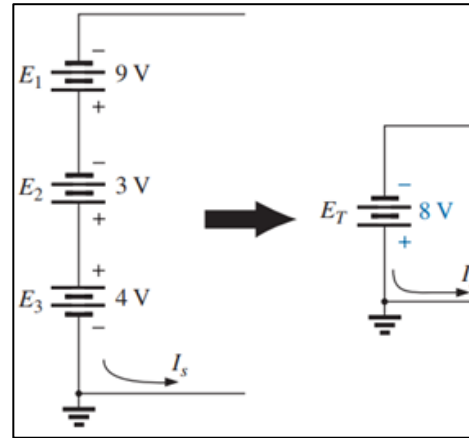
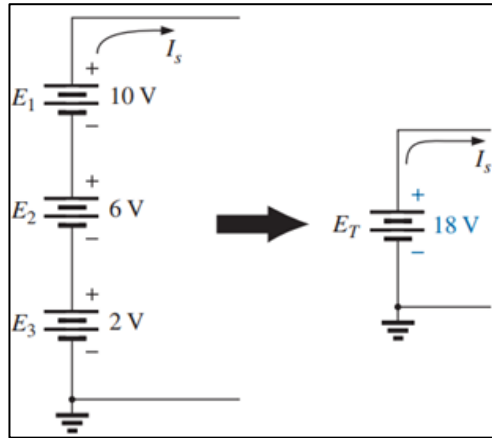
- Using circuit laws, determine  $I_o$ . Also determine the power of the  $2I_o$  source.



Ans:  $I_o = -3 \text{ A}$ ;  $P_{2I_o} = -810 \text{ W}$



# Series and Parallel sources

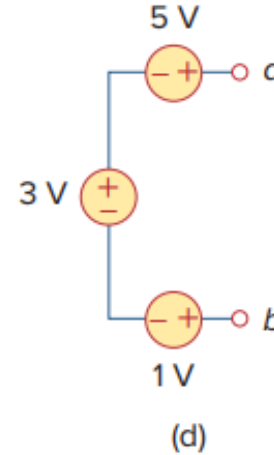
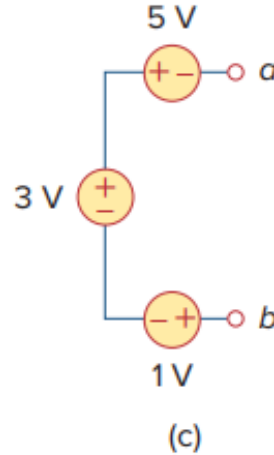
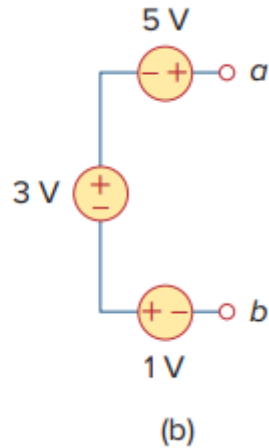
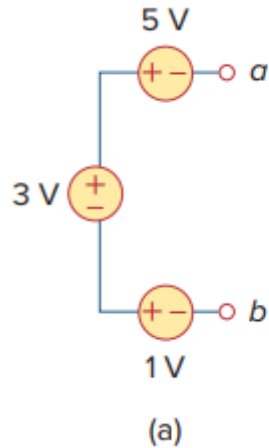


It is not practical to connect voltage sources of unequal ratings in parallel and current sources of unequal currents in series due to the direct violation of KVL and KCL respectively.



# Problem 15

- For each of the circuits shown below, calculate  $V_{ab}$

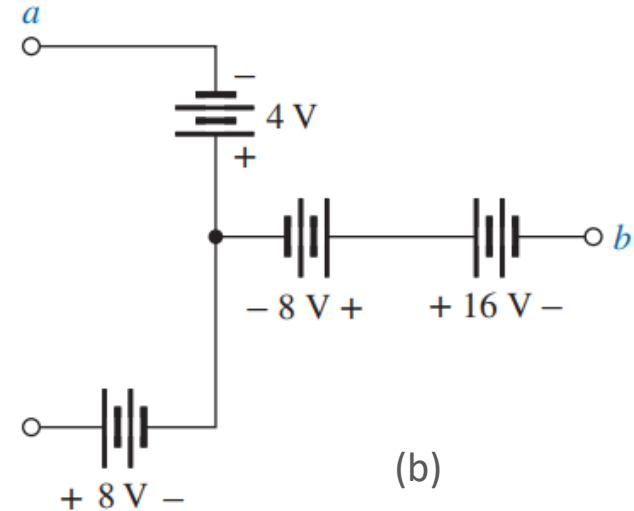
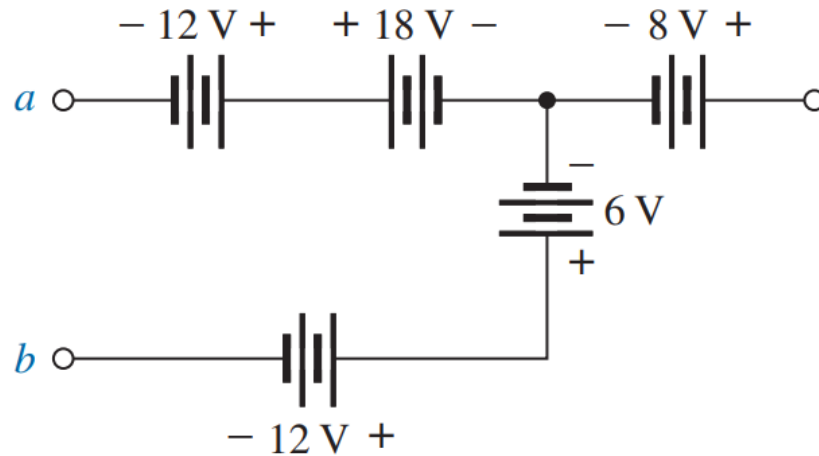


Ans: (a)  $V_{ab} = -1\text{ V}$ ; (b)  $V_{ab} = 9\text{ V}$ ; (c)  $V_{ab} = -3\text{ V}$ ; (d)  $V_{ab} = 7\text{ V}$



# Problem 16

- For each of the circuits shown below, calculate  $V_{ab}$ .

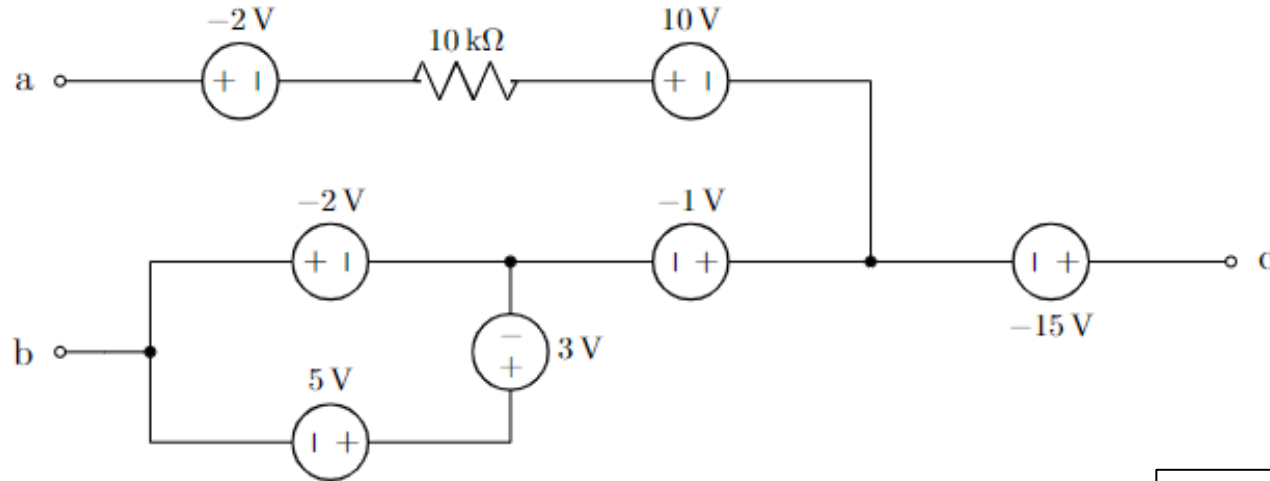


Ans: (a)  $V_{ab} = 12\text{ V}$ ; (b)  $V_{ab} = 4\text{ V}$



# Problem 17

- For the circuit shown below, calculate  $V_{ac}$  and  $V_{bc}$



Ans:  $V_{ac} = 23\text{ V}$ ;  $V_{bc} = 14\text{ V}$



# Voltage Division Rule

- The voltage division rule permits the determination of the voltage across a series resistor without first having to determine the current of the circuit.

- The current through the series circuit can be found using Ohm's law as,

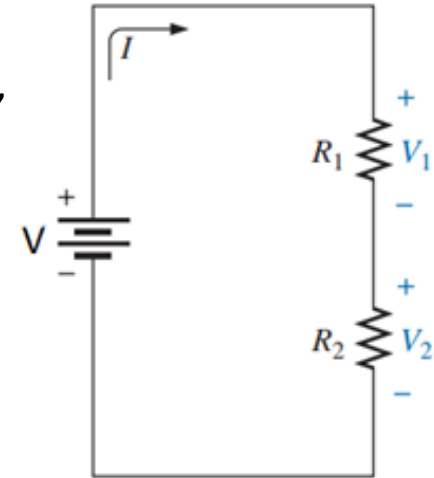
$$I = \frac{V}{R_1 + R_2}$$

- Applying Ohm's law to each of the resistors,

$$V_1 = IR_1 \quad \text{and} \quad V_2 = IR_2$$

$$\Rightarrow V_1 = \frac{V}{R_1 + R_2} R_1 \quad \text{and} \quad V_2 = \frac{V}{R_1 + R_2} R_2$$

$$\Rightarrow V_1 = \frac{R_1}{R_1 + R_2} \times V \quad \text{and} \quad V_2 = \frac{R_2}{R_1 + R_2} \times V$$



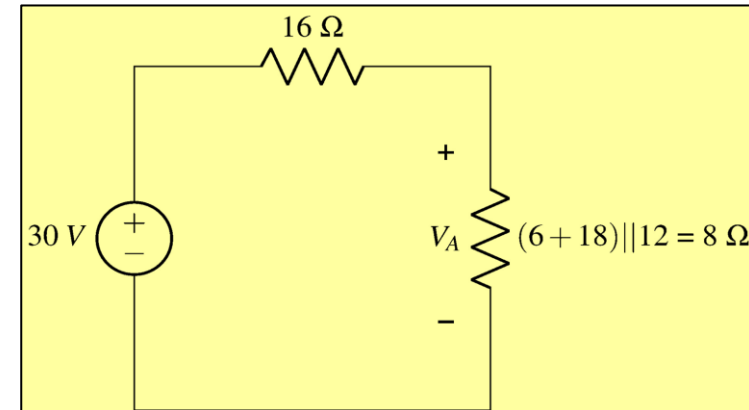
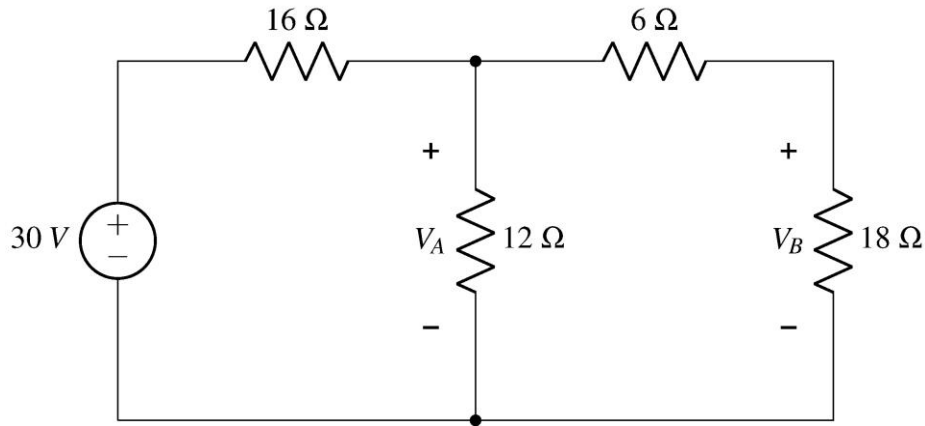
- In general, for any number of resistors connected in series to a supply voltage, the voltage across any particular resistor  $R_x$  is,

$$V_x = \frac{R_x}{R_1 + R_2 + R_3 + \dots + R_N} \times V$$



# Example 5

- Using the voltage divider rule, find the voltages  $V_A$  and  $V_B$ . Don't calculate currents.



$$V_A = \frac{8}{8 + 16} \times 30 = 10 \text{ V}$$

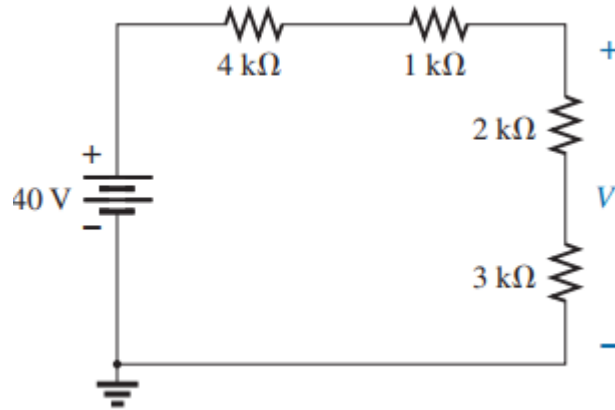


$$V_B = \frac{18}{18 + 6} \times V_A = 7.5 \text{ V}$$



# Problem 18

- Using the voltage divider rule, find the indicated voltage. Don't calculate current.

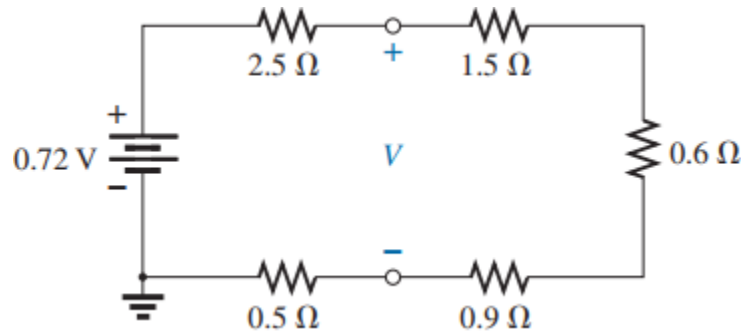


Ans:  $V = 20V$



# Problem 19

- Using the voltage divider rule, find the indicated voltage. Don't calculate current.

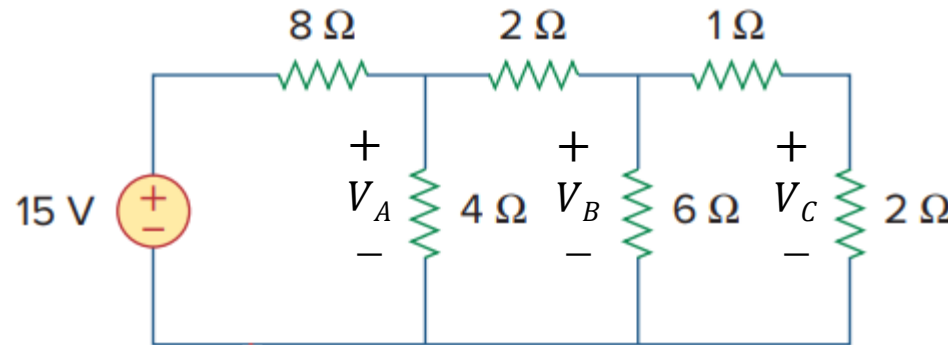


$$\text{Ans: } V = 0.36 \text{ V}$$



# Problem 20

- Using the voltage divider rule, find the voltages  $V_A$ ,  $V_B$ , and  $V_C$ . Don't calculate currents.

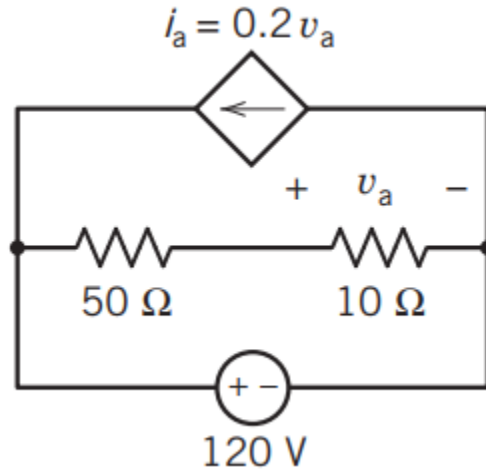


**Ans:  $V_A = 3\text{ V}$ ;  $V_B = 1.5\text{ V}$ ;  $V_C = 1\text{ V}$**



# Problem 21

- Determine the power of the dependent source. Don't use Ohm's Law.

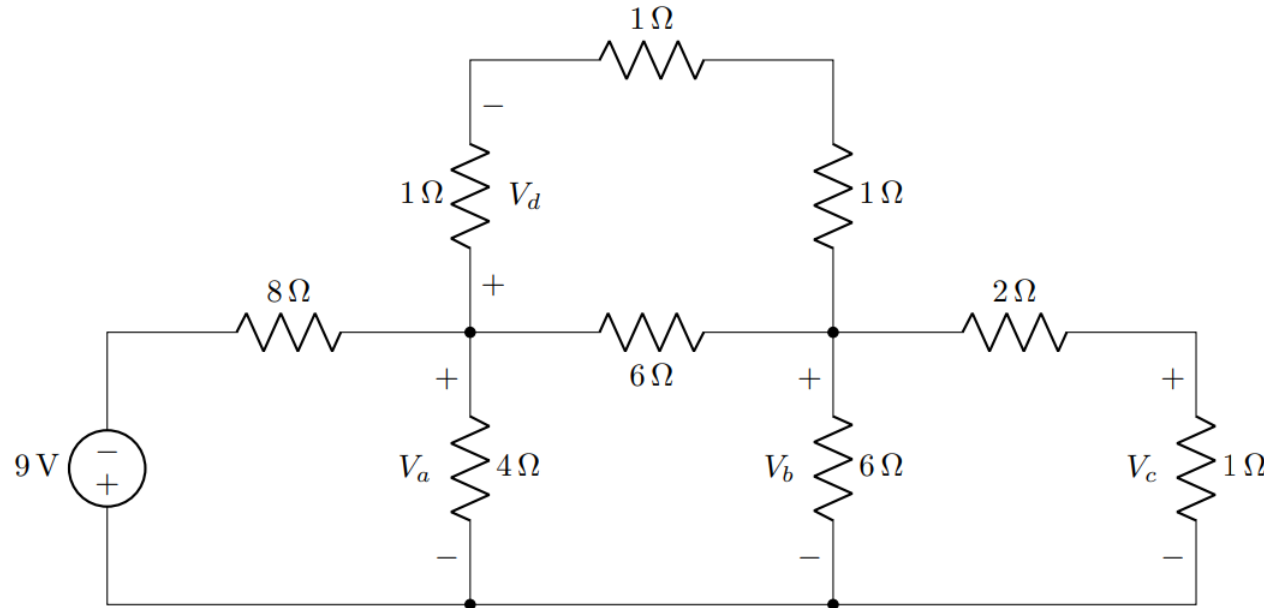


**Ans:  $-480\text{ W}$**



# Problem 22

- Determine  $V_a$ ,  $V_b$ ,  $V_c$ , and  $V_d$  using voltage division rule. Don't use Ohm's Law.



**Ans:**  $V_a = -1.8\text{ V}$ ,  $V_b = -0.9\text{ V}$ ,  $V_c = -0.3\text{ V}$ ,  $V_d = -0.3\text{ V}$



# Current Division Rule

- The current division rule permits the determination of the currents through resistors connected in parallel without first having to determine the voltage across them.

- Since the voltage  $V$  is the same across parallel elements, the following is true:

$$V = I_1 R_1 = I_2 R_2 = I_3 R_3 = \dots = I_N R_N$$

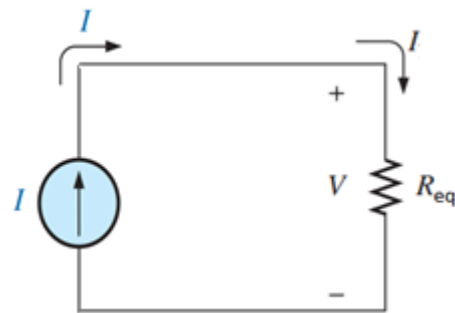
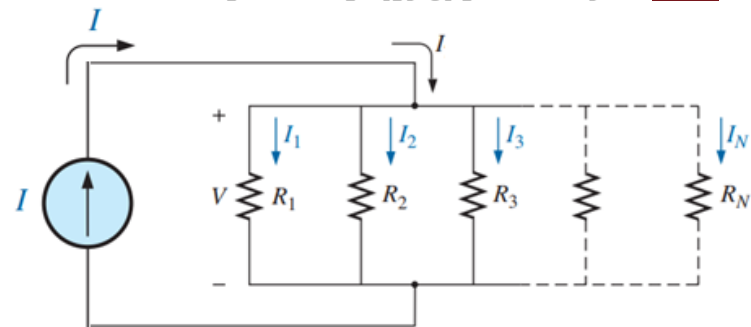
- Substituting  $V$  with  $V = I R_{eq}$ ,

$$I R_{eq} = I_1 R_1 = I_2 R_2 = I_3 R_3 = \dots = I_N R_N$$

$$\Rightarrow I_1 = \frac{R_{eq}}{R_1} \times I, \quad I_2 = \frac{R_{eq}}{R_2} \times I, \quad I_3 = \frac{R_{eq}}{R_3} \times I$$

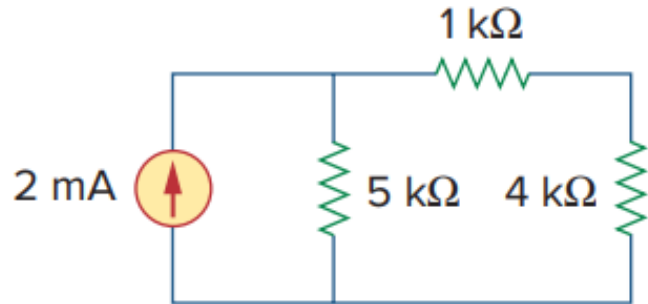
- In general, for any number of resistors connected in parallel to a supply current, the current through any particular resistor  $R_x$  is,

$$I_x = \frac{R_{eq}}{R_x} \times I, \text{ or, } I_x = \frac{(R_x)^{-1}}{(R_1)^{-1} + (R_2)^{-1} + \dots + (R_N)^{-1}} \times I$$



# Example 6

- Calculate the current through the  $5\text{ k}\Omega$  resistor using current division rule. Do not use Ohm's Law.



## Solution

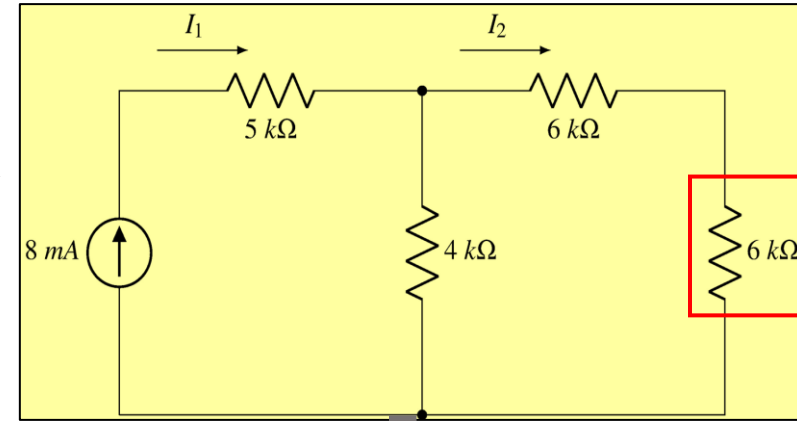
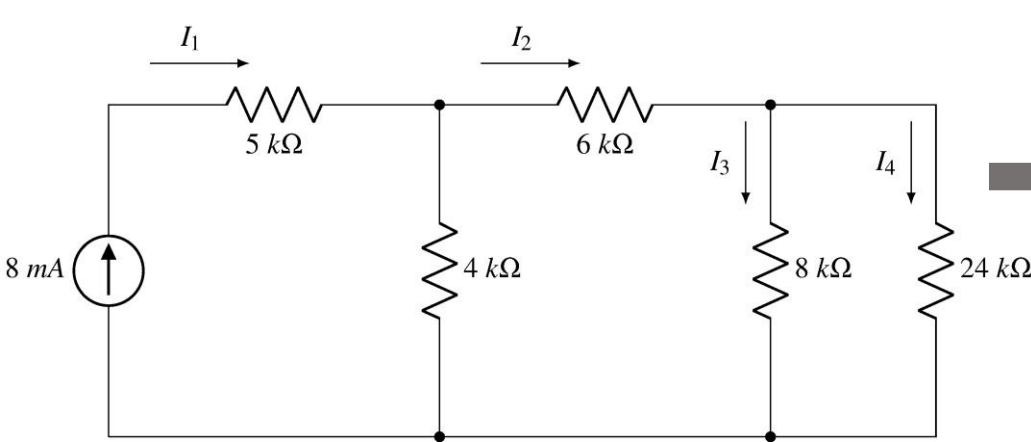
Current through the  $5\text{ k}\Omega$  resistor is,

$$\frac{5^{-1}}{(1 + 4)^{-1} + 5^{-1}} \times 2\text{ mA} = 1\text{ mA}$$



# Example 7

- Calculate the currents  $I_1$  to  $I_4$  using current division rule. Don't calculate voltage.



$$I_3 = \frac{8^{-1}}{24^{-1} + 8^{-1}} \times I_2 = 1.5 \text{ mA}$$

$$I_4 = I_2 - I_3 = 0.5 \text{ mA}$$

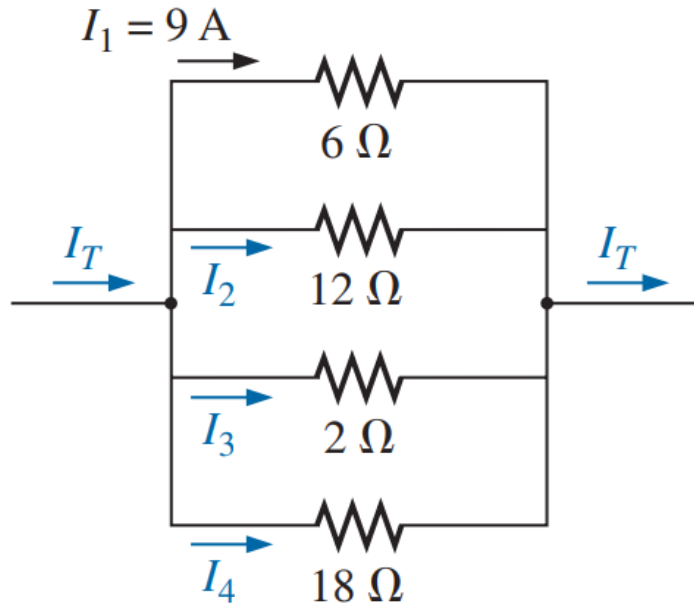
$$I_1 = 8 \text{ mA}$$

$$I_2 = \frac{12^{-1}}{4^{-1} + 12^{-1}} \times 8 \text{ mA} = 2 \text{ mA}$$



# Problem 23

- Based solely on the resistor values, determine all the currents. Do not use Ohm's law.

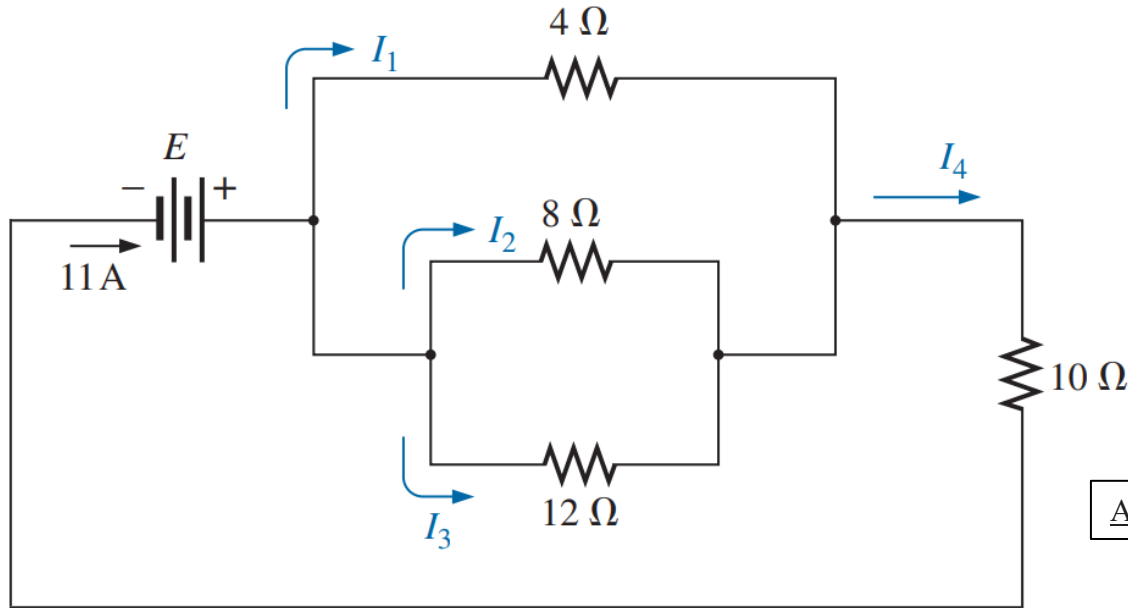


Ans:  $I_T = 43.5\text{ A}$ ;  $I_2 = 4.5\text{ A}$ ;  $I_3 = 27\text{ A}$ ;  $I_4 = 3\text{ A}$



# Problem 24

- Determine the unknown currents. Do not use Ohm's law.

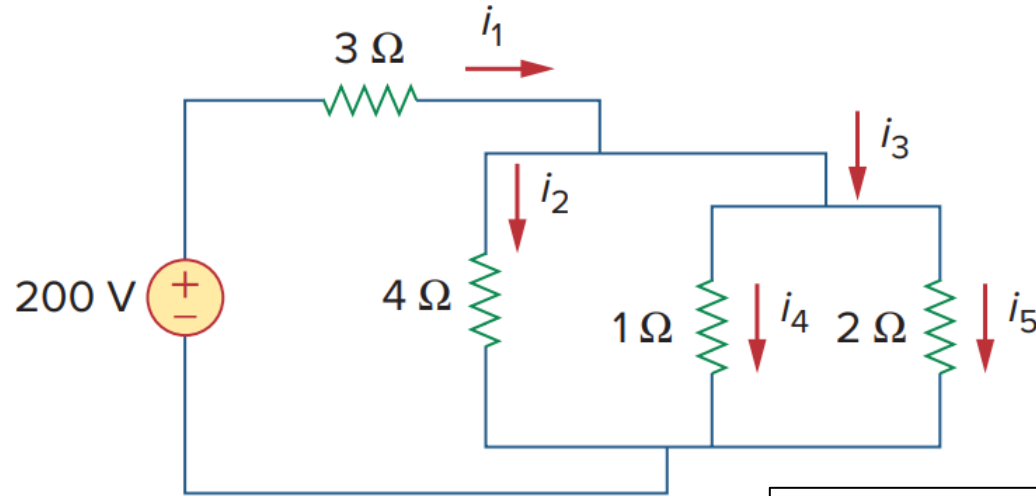


Ans:  $I_1 = 6\text{ A}$ ;  $I_2 = 3\text{ A}$ ;  $I_3 = 2\text{ A}$ ;  $I_4 = 11\text{ A}$



# Problem 25

- Determine the currents  $i_1$  to  $i_5$  using current division rule.



Ans:  $i_1 = 56\text{ A}$ ;  $i_2 = 8\text{ A}$ ;  $i_3 = 48\text{ A}$ ;  $i_4 = 32\text{ A}$ ;  $i_5 = 16\text{ A}$ .



# Additional Problems

- Additional recommended practice problems: [here](#)
- Other suggested problems from the textbook: [here](#)

# Acknowledgements and References

- Some of the problems, illustrations, and concepts in this lecture are taken from the following sources:
  1. Sadiku, M. N. O., Fundamentals of Electric Circuits, McGraw-Hill
  2. Dorf, R. C., & Svoboda, J. A., Introduction to Electric Circuits, Wiley
  3. Boylestad, R. L., Introductory Circuit Analysis, Pearson



# Thank you for your attention

